

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1565

Orchard Road 138 kV
April 02, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1565. The Interconnection Customer proposes a 270.0 MW solar photovoltaic facility interconnecting at the Orchard Road 138 kV substation in the Pennsylvania area. The request was received on September 07, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 481 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,292,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Talenenergy Martins Creek Llc Lock Haven-El Cerrito Sta G 138 kV
Contingency: Scott-American Crystal Sugar Drayton 345 kV line
Loading: 116.2% (Rating: 235 MVA)
DFAX: 5.2%
Pre-project loading: 110.3%
2. Facility: Talenenergy Martins Crk Llc Lock Haven-Taylor Flats 138 kV
Contingency: American Crystal Sugar Drayton-Scott 345 kV L
Loading: 103.0% (Rating: 265 MVA)
DFAX: 5.2%
Pre-project loading: 97.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Talenenergy Martins Crk Llc Lock Haven-El Cerrito Sta G Substation 138 kV	Rebuild line with higher-capacity conductor	\$9,084,000	19% (\$1,694,000)
Talenenergy Martins Crk Llc Lock Haven-Taylor Flats 138 kV	Reconductor line	\$3,206,000	19% (\$598,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$2,292,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.